

Path Dependence in a Shared-State Activity Poll

Ten repeated EDSL simulations with sequentially visible votes

Main result. The same balanced preference distribution produced four different winners. Winner frequency was bike ride: 3, sailing: 3, hike: 3, beach day: 1. Once an early lead formed, later agents' desire to join the group often amplified it into a large majority.

Research question

Can sequentially visible choices produce different collective outcomes when the group's underlying activity preferences begin balanced? This demonstration repeats the same shared-state survey ten times. It is designed to make the mechanism observable: every agent reads the current vote register, answers once, and writes one vote.

Experimental design

Each run contains 16 simulated participants and four activities: *bike ride*, *sailing*, *hike*, and *beach day*. Exactly four participants prefer each activity before ordering. Each participant also receives:

- a preference strength drawn from 0.60 to 0.90; and
- a conformity value drawn from 0.55 to 0.90.

The ordering and trait values vary with the run seed. Agents are prompted through Gemini 2.5 Flash using local EDSL orchestration. A grouped round-robin schedule creates an ordering barrier, so voter t sees the state committed by voters $1, \dots, t - 1$. Across the ten runs there are 160 decisions.

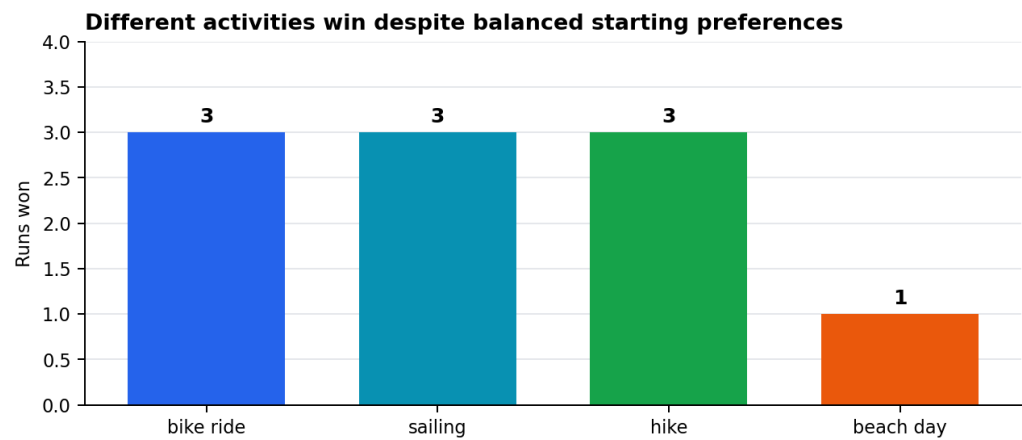
Shared-state mechanism

For each participant, the survey executes the following sequence:

1. Read the vote register at version $t - 1 \rightarrow$ **2. Ask** the activity question with that snapshot $\rightarrow$ **3. Write** the selected activity $\rightarrow$ **4. Commit** version t .

The explicit read is important. It records what the participant could observe and makes the prompt's state dependency auditable in the Results provenance. Concurrency limits alone would not create this sequence; the schedule supplies the barrier between turns.

Outcomes across repetitions



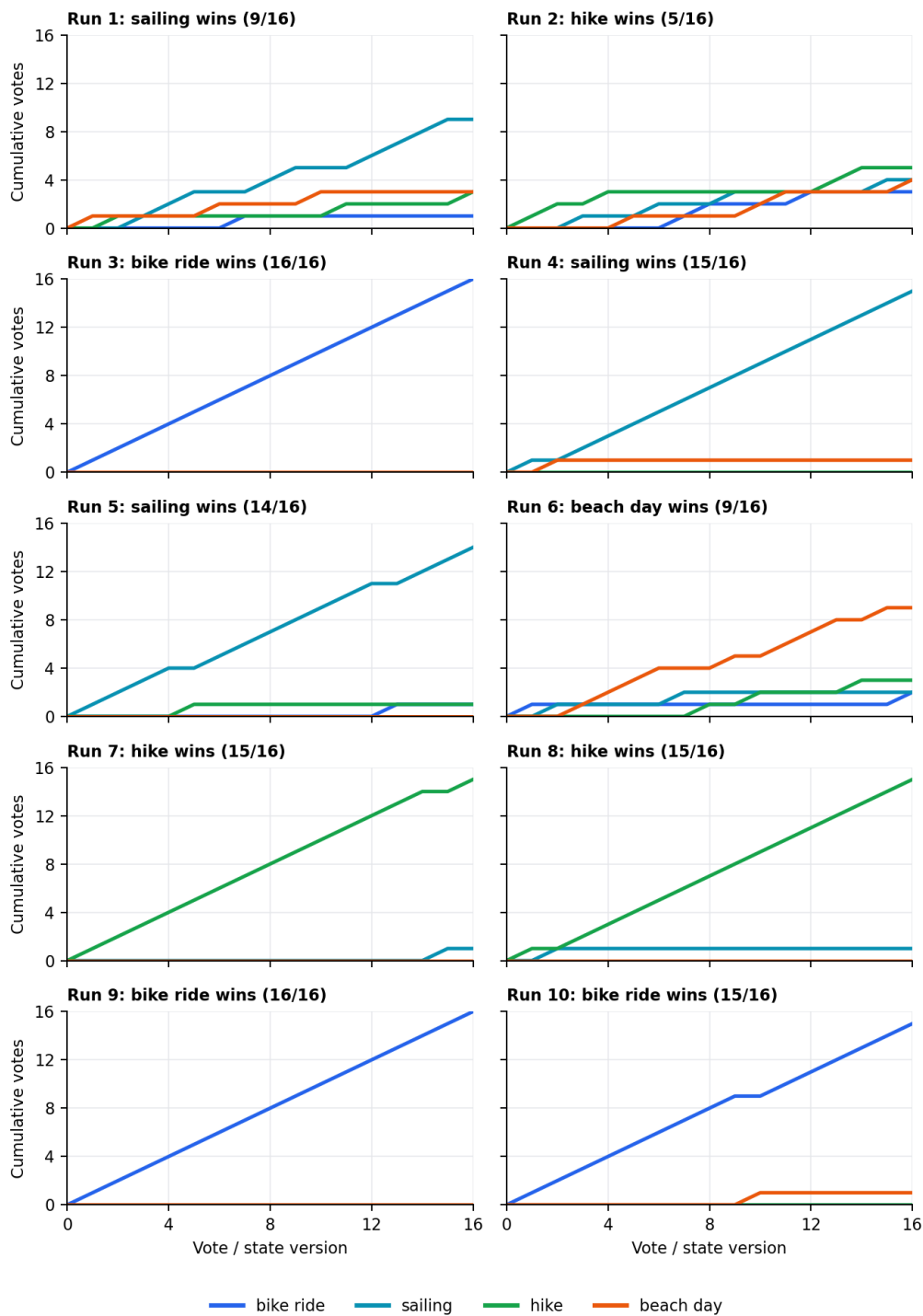
The mean winning total was 12.9 of 16 votes. Participants selected their initially preferred activity in 44.4choices provide a direct descriptive measure of departures from baseline preference.

Run	First three votes	Bike	Sail	Hike	Beach	Winner
1	beach day, hike, sail- ing	1	9	3	3	sailing
2	hike, hike, sailing	3	4	5	4	hike
3	bike ride, bike ride, bike ride	16	0	0	0	bike ride
4	sailing, beach day, sailing	0	15	0	1	sailing
5	sailing, sailing, sail- ing	1	14	1	0	sailing
6	bike ride, sailing, beach day	2	2	3	9	beach day
7	hike, hike, hike	0	1	15	0	hike
8	hike, sailing, hike	0	1	15	0	hike
9	bike ride, bike ride, bike ride	16	0	0	0	bike ride
10	bike ride, bike ride, bike ride	15	0	0	1	bike ride

Table 1. Final counts and the earliest observed path. Preferences were balanced 4-4-4-4 in every run.

How the vote paths evolved

The ten realized vote paths



What the repetitions show

Multiple outcomes occurred. Bike ride, sailing, hike, and beach day each won at least once, even though every run began with four agents assigned to prefer each activity.

Early positions were informative. The eventual winner was among the activity or activities tied for the lead after three votes in 10 of 10 runs. This wording matters: runs with three different opening votes have a three-way early tie, not a unique predictor.

Some paths cascaded. Runs 3 and 9 ended 16-0; runs 4, 7, 8, and 10 ended 15-1. The line plots expose when those gaps opened and whether later voters reinforced them.

Interpretation. The simulations are consistent with a path-dependent process: early random variation changes the public state, and the public state can influence later choices. They do not, by themselves, estimate a causal conformity effect. Both the trait draws and model sampling vary across runs.

Limitations and next tests

- Ten repetitions are illustrative rather than a stable estimate of outcome probabilities.
- The agents are language-model simulations, not human participants.
- Ordering, trait values, and model stochasticity vary together in this version.
- A stronger test would hold personas fixed, randomly permute only their order, and run many more repetitions.
- A no-visibility control would separate preference-driven choices from responses to the live vote register.

Reproducibility

The report is generated from `shared-state-activity-poll-repeated.json`. The single-run implementation is `shared_state_activity_poll.py`, the ten-run driver is `shared_state_activity_poll_repeated.py`, and the report builder is `build_shared_state_path_dependence_report.py`.